

Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.6b)

Metric

$$ds^2 = -dt \otimes dt + \frac{3}{4} \cdot 3^{\frac{1}{3}} 2^{\frac{2}{3}} m^{\frac{2}{3}} t^{\frac{4}{3}} dr \otimes dr + \frac{3}{4} \cdot 3^{\frac{1}{3}} 2^{\frac{2}{3}} m^{\frac{2}{3}} r^2 t^{\frac{4}{3}} d\theta \otimes d\theta + \frac{3}{4} \cdot 3^{\frac{1}{3}} 2^{\frac{2}{3}} m^{\frac{2}{3}} r^2 t^{\frac{4}{3}} \sin^2(\theta) d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{1}{2} \sqrt{2} \left(\frac{3}{2} \right)^{\frac{2}{3}} (\sqrt{mt})^{\frac{2}{3}} \right) dr$$

$$n = \left(\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{1}{2} \sqrt{2} \left(\frac{3}{2} \right)^{\frac{2}{3}} (\sqrt{mt})^{\frac{2}{3}} \right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} \left(\frac{3}{2} \right)^{\frac{2}{3}} (\sqrt{mt})^{\frac{2}{3}} r \right) d\theta + \left(\frac{1}{2} i \sqrt{2} \left(\frac{3}{2} \right)^{\frac{2}{3}} (\sqrt{mt})^{\frac{2}{3}} r \sin(\theta) \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \left(\frac{3}{2} \right)^{\frac{2}{3}} (\sqrt{mt})^{\frac{2}{3}} r \right) d\theta + \left(-\frac{1}{2} i \sqrt{2} \left(\frac{3}{2} \right)^{\frac{2}{3}} (\sqrt{mt})^{\frac{2}{3}} r \sin(\theta) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} t^{\frac{2}{3}}} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} t^{\frac{2}{3}}} \right) dr$$

$$m = \left(\frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}}} \right) d\theta + \left(\frac{i \cdot 3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}} \sin(\theta)} \right) d\phi$$

$$\overline{m} = \left(\frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}}} \right) d\theta + \left(-\frac{i \cdot 3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}} \sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} t^{\frac{2}{3}}} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t - \frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} t^{\frac{2}{3}}} \partial_r$$

$$\delta = \frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}}} \partial_\theta + \frac{i \cdot 3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}} \sin(\theta)} \partial_\phi$$

$$\overline{\delta} = \frac{3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}}} \partial_\theta - \frac{i \cdot 3^{\frac{1}{3}} 2^{\frac{1}{6}}}{3 m^{\frac{1}{3}} r t^{\frac{2}{3}} \sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}}{3t} - \frac{3^{\frac{1}{3}}2^{\frac{1}{6}}}{3m^{\frac{1}{3}}rt^{\frac{2}{3}}}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}}{3t} - \frac{3^{\frac{1}{3}}2^{\frac{1}{6}}}{3m^{\frac{1}{3}}rt^{\frac{2}{3}}}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{3^{\frac{1}{3}}2^{\frac{1}{6}}\cos(\theta)}{6m^{\frac{1}{3}}rt^{\frac{2}{3}}\sin(\theta)}$$

$$\beta = \frac{3^{\frac{1}{3}}2^{\frac{1}{6}}\cos(\theta)}{6m^{\frac{1}{3}}rt^{\frac{2}{3}}\sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}}{6t}$$

$$\gamma = \frac{\sqrt{2}}{6t}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{1}{3t^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{6t^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{3t^2}$$

$$\Lambda = \frac{1}{18t^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"